

SEDIGHE RAEISI, PH.D.

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SKILLS & EXPERTISE

- **Programming:** Python, Fortran, C#, LaTeX.
- **Machine Learning:** Probabilistic AI, AI4Science, Computer Vision, NLP, Reinforcement Learning, Time Series.
- **Deep Learning Libraries:** PyTorch, TensorFlow, Jax, NumPyro.
- **Neural Networks:** GCN, LSTM, RNN, CNN, ANN.
- **Mathematical Computing:** Maple, Mathematica.
- **Softwares & Databases:** Neo4j, Unreal Engine, Flask, Docker, Git, Origin, LyX, Photoshop.

WORK EXPERIENCE

- Universität Osnabrück** Osnabrück, Germany
Alexander von Humboldt Postdoctoral Researcher 2024 – Present
- **Project Leader:** Developed **Hierarchical Bayesian** identification of non-linear dynamics for model discovery.
 - Conducted data-driven discovery of individual differences in cognitive mechanisms using **HPC** computations.
 - Implemented Knowledge Graphs with **Neo4j** in automated laboratories for integrated scientific discovery.
 - Professional **LLM** Development: Optimized research workflows by pre-training a **Qwen-code** model on a specialized **Hierarchical Bayesian SINDy** GitHub repo; successfully deployed the model to **Hugging Face** and integrated it into a local **Ollama** environment for advanced, automated equation discovery.
 - Time-Series Forecasting with Foundational Models.
 - **Co-Supervisor:** Data-driven model discovery with SINDy for climate data with collaboration with the University of New South Wales, Sydney, Australia.
- MCI R&D Center (Hamrah-e-Aval)** Tehran, Iran
Data Scientist 2022 – 2024
- Designed and implemented a physics-informed Graph Neural Network (GNN) for PCI planning for the first time.
 - Utilized Pytorch GCN, NetworkX, Geopandas, and Folium for spatial network analysis and planning.
 - Developed RAG (Retrieval-Augmented Generation) systems using Langchain and ChromaDB for NLP tasks.
 - Conducted Time-Series Forecasting using Arima, Sarima, RNN, LSTM, and Facebook Prophet.
 - Created a Live AI Hologram using Unreal Engine, featuring lip-sync, text-to-speech, and animation.
 - Alarm Prediction with Hierarchical classification and graph neural network, improved accuracy from 40% to 80%.

PROJECTS

- **Computer Vision:**
 - **Brain Tumor Detection:** Used a CNN classifier and VGG16 trained with augmented MRI images to predict tumor existence.
 - **Tumor Localization:** Trained Mask-RCNN for tumor masking and YOLO-V5 for bounding box detection.
 - **Persian Digit Recognition:** Developed image preprocessing and detection using OpenCV, SkImage, CNN, and Pytesseract.
- **NLP & Sentiment Analysis:**
 - **Stock Market:** Content analysis of financial Telegram channels using Pretrained embeddings, fine-tuned BERT, and TF-IDF.
 - **Protein Analysis:** Analyzed protein molecule series using NLP techniques in collaboration with a German research group.
 - **Digikala:** Sentiment prediction of customer posts using the Hazm library for Persian text preprocessing.
- **Graph Neural Networks:** Executed graph classification on MNIST, node classification on Karate Club, and edge prediction.
- **Clustering:** Segmented bank customers using K-means and DBSCAN based on transaction and balance history.
- **Scientific Projects:** Investigated particle behavior in potential wells with classic chaotic behavior using the Boundary Integral method, using physical simulation, numerical methods, and the finite element method.

TEACHING EXPERIENCE

- Lecturer @ Ferdowsi University of Mashhad** 2017 – 2020
- **Physics 1 & 2:** Departments of Industrial, Civil, Electrical, Agriculture, and Mechanical Engineering.
 - **General Physics:** Biology department.

- **Lab Instructor:** Led Physics Lab 1 & 2 for Physics, Chemistry, Agriculture, and Electrical Engineering students.
- **Teaching Assistant:** Courses included Computational Physics (Graduate level), Computer Programming (Fortran), Modern Physics, and Physics 2.

Additional Instructor Roles

- **"Machine Learning with Python" & "Deep Learning with Python":** Instructor for the Mooc of Tehran University.
- **Python OOP:** Taught programming at Mehr-e-madar and Pishkesvatan Institutes.
- **"Teaching High School Mathematics with Maple":** Instructor at Faradars Institute.

EDUCATION

Ferdowsi University of Mashhad, Faculty of Science	Mashhad, Iran
Ph.D. in Physics (Elementary Particles)	2012 – 2017
<i>Thesis: Investigation of Fermi Acceleration in a Non-planar Time-Dependent Billiard</i>	
<i>Supervisor: Dr. P. Eslami</i>	
M.Sc. in Physics (Elementary Particles)	2010 – 2012
<i>Thesis: Quantum chaos in Kicked Particles by Gaussian Potential</i>	
<i>Supervisors: Dr. P. Eslami, Dr. Mohsen Sarbishaei</i>	
B.Sc. in Physics	2005 – 2009

PUBLICATIONS & PRESENTATIONS

- [Machine Learning-Based Equation Discovery in Land-Atmosphere Modelling](#), *2026 Pan-GLASS Conference*, 2026 (Accepted for presentation, Stuttgart, Germany; **Co-Supervisor & Presenter**).
- [Computational discovery of individual differences in cognitive mechanisms](#), *Dynamics Days Europe*, 2025 (Talk).
- [Computational discovery of individual differences in cognitive mechanisms](#), *KogWis*, 2025 (Talk).
- [Automated discovery of structural differences in dynamical systems](#), *AI in Science Summit*, 2025 (Poster).
- [PCI planning with graph neural network](#), *8th SCIoT IEEE*, 2024.
- [Machine Learning for Physicists](#), **IOP Publishing Limited**, 2023.
- [Acceleration in a nonplanar time-dependent billiard](#), *Phys. Rev. E*, 2016.
- Study of energy growth in non-Euclidean time-dependent geometries, *Annual Physics Conference of Iran*, 2015.

HONORS AND AWARDS

- **Alexander von Humboldt Postdoctoral Researcher Fellowship**, 2024.
- **Ranked first** among Ph.D. students of the Class of 2012, Ferdowsi University of Mashhad.
- **Ranked second** in the Scientific Poster Competition, Faculty of Science, Ferdowsi University of Mashhad.
- **Selection of work** for publication in the Student Achievement Festival Book.